

Ex.No.2

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**DATA MANIPULATION LANGUAGE, DATA CONTROL
LANGUAGE COMMANDS AND TCL COMMANDS**

AIM

To write and execute Data manipulation language, data control language and transfer control logic commands in SQL.

Data Manipulation Language

1. Creation & inserting values into Table

```
SQL> CREATE TABLE library(book_no NUMBER(5) PRIMARY KEY, title VARCHAR2(10) NOT NULL UNIQUE,  
author VARCHAR2(10), genre VARCHAR(15) CHECK (genre IN ('fiction', 'non-fiction',  
'science'))), pages NUMBER(3));
```

Table created.

Inserting Values in the Table

Inserting Values With Selected Attributes

```
SQL> INSERT INTO library(book_no, title, genre, pages) VALUES (101, 'To Kill a Mockingbird',  
'fiction', 281);
```

1 row created.

```
SQL> INSERT INTO library(book_no, title, genre, pages) VALUES (102, 'The Great Gatsby',  
'fiction', 180);
```

1 row created.

System Input

```
SQL> INSERT INTO library VALUES (&book_no, '&title', '&author', '&genre', &pages);
```

Enter value for book_no: 103

Enter value for title: War and Peace

Enter value for author: Leo Tolstoy

Enter value for genre: fiction

Enter value for pages: 122

```
old 1: INSERT INTO library VALUES (&book_no, '&title', '&author', '&genre', &pages)
```

```
new 1: INSERT INTO library VALUES (103, 'War and Peace', 'Leo Tolstoy', 'fiction', 122)
```

1 row created.

```
SQL> /
```

Enter value for book_no: 104

Enter value for title: The Odyssey

Enter value for author: Homer

Enter value for genre: fiction

Enter value for pages: 400

```
old 1: INSERT INTO library VALUES (&book_no, '&title', '&author', '&genre', &pages)
```

```
new 1: INSERT INTO library VALUES (104, 'The Odyssey', 'Homer', 'fiction', 400)
```

1 row created.

Manual Input

```
SQL> INSERT INTO library VALUES (105, 'The Guide', 'R.K. Narayan', 'fiction', 220);
```

1 row created.

```
SQL> INSERT INTO library VALUES (106, 'WingsFire', 'Kalam', 'science', 180);
```

1 row created.

```
SQL> INSERT INTO library VALUES (107, 'ThinkFast', 'Kahneman', 'non-fiction', 499);
1 row created.
```

TO SEE THE STRUCTURE OF THE TABLE:

```
SQL> desc library;
```

Name	Null?	Type
BOOK_NO	NOT NULL	NUMBER(3)
TITLE	NOT NULL	VARCHAR2(30)
AUTHOR		VARCHAR2(20)
GENRE		VARCHAR2(15)
PAGES		NUMBER(3)

TO VIEW ENTIRE TABLE:

SQL> select * from library:

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird		fiction	281
102	The Great Gatsby		fiction	180
103	War and Peace	Leo Tolstoy	fiction	122
104	The Odyssey	Homer	fiction	400
105	The Guide	R.K. Narayan	fiction	220
106	WingsFire	Kalam	science	180
107	ThinkFast	Kahneman	non-fiction	499
108	Sapiens	Yuval Noah	non-fiction	522

8 rows selected.

Updating values in the table manually without case:

```
SQL> update library set pages = pages + 20 where genre = 'fiction';
```

5 rows updated.

```
SQL> update library set pages = pages - 10 where genre = 'non-fiction';
```

2 rows updated.

```
SQL> update library set pages = pages + 30 where genre = 'science';
```

1 row updated.

```
SQL> select * from library;
```

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird		fiction	301
102	The Great Gatsby		fiction	200
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	512

8 rows selected.

Updating with Case keyword:

```
SQL> update library set pages = case
2  when title = 'To Kill a Mockingbird' then pages + 10
```

```

3  when author = 'Yuval Noah' then pages - 10
4  else pages
5  end;

```

8 rows updated.

SQL> select * from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	Leo Tolstoy	fiction	311
102	The Great Gatsby		fiction	200
103	War and Peace		fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

8 rows selected.

Select Command :

a.Selecting Column:

SQL> select title, author from library;

TITLE	AUTHOR
To Kill a Mockingbird	Leo Tolstoy
The Great Gatsby	
War and Peace	
The Odyssey	Homer
The Guide	R.K. Narayan
WingsFire	Kalam
ThinkFast	Kahneman
Sapiens	Yuval Noah

8 rows selected.

b. selecting rows:

SQL> select * from library where genre = 'non-fiction';

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

c.Selecting all the rows:

SQL> select * from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	Harper Lee	fiction	311
102	The Great Gatsby	F. Scott Fitzgerald	fiction	200
103	War and Peace	Leo Tolstoy	fiction	142

104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	A.P.J. Abdul Kalam	science	210
107	ThinkFast	Daniel Kahneman	non-fiction	489
108	Sapiens	Yuval Noah Harari	non-fiction	502

8 rows selected.

d.Select Clause With Arithmetic And Logical Operators :

Arithmetic Operator:

SQL> select title, pages+5 from library;

TITLE	PAGES+5
-----	-----
To Kill a Mockingbird	316
The Great Gatsby	205
War and Peace	147
The Odyssey	425
The Guide	245
WingsFire	215
ThinkFast	494
Sapiens	507

8 rows selected.

SQL> select title, pages*2 from library;

TITLE	PAGES*2
-----	-----
To Kill a Mockingbird	622
The Great Gatsby	400
War and Peace	284
The Odyssey	840
The Guide	480
WingsFire	420
ThinkFast	978
Sapiens	1004

8 rows selected.

SQL> select title, pages-10 from library;

TITLE	PAGES-10
-----	-----
To Kill a Mockingbird	301
The Great Gatsby	190
War and Peace	132
The Odyssey	410
The Guide	230
WingsFire	200
ThinkFast	479
Sapiens	492

8 rows selected.

Logical Operator:

SQL> select title, genre, pages from library where pages > 200;

TITLE	GENRE	PAGES
-----	-----	-----
To Kill a Mockingbird	fiction	311
The Odyssey	fiction	420
The Guide	fiction	240
WingsFire	science	210
ThinkFast	non-fiction	489
Sapiens	non-fiction	502

6 rows selected.

SQL> select * from library where pages >= 240;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	Harper Lee	fiction	311
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
107	ThinkFast	Daniel Kahneman	non-fiction	489
108	Sapiens	Yuval Noah Harari	non-fiction	502

SQL> select * from library where pages <= 180;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
103	War and Peace	Leo Tolstoy	fiction	142

SQL> select * from library where genre = 'fiction';

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	Harper Lee	fiction	311
102	The Great Gatsby	F. Scott Fitzgerald	fiction	200
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240

SQL> select * from library where genre != 'non-fiction';

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	Harper Lee	fiction	311
102	The Great Gatsby	F. Scott Fitzgerald	fiction	200
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	Wings of Fire	A.P.J. Abdul Kalam	science	210

SQL> select * from library where genre = 'fiction' and pages > 200;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	Harper Lee	fiction	311
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240

SQL> select * from library where pages between 200 and 500;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	Harper Lee	fiction	311
102	The Great Gatsby	F. Scott Fitzgerald	fiction	200

104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	Wings of Fire	A.P.J. Abdul Kalam	science	210
107	ThinkFast	Daniel Kahneman	non-fiction	489

e.Select with <tablename> . <attribute> command:

SQL> select library.* from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
-----	-----	-----	-----	-----
101	To Kill a Mockingbird	-	fiction	311
102	The Great Gatsby	-	fiction	200
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	Wings of Fire	Kalam	science	210
107	Think Fast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

8 rows selected.

SQL> select library.book_no, library.title from library;

BOOK_NO	TITLE
-----	-----
101	To Kill a Mockingbird
102	The Great Gatsby
103	War and Peace
104	The Odyssey
105	The Guide
106	WingsFire
107	ThinkFast
108	Sapiens

8 rows selected.

Select with alias:

SQL> select l.title, l.author from library l;

TITLE	AUTHOR
-----	-----
To Kill a Mockingbird	
The Great Gatsby	
War and Peace	Leo Tolstoy
The Odyssey	Homer
The Guide	R.K. Narayan
WingsFire	Kalam
ThinkFast	Kahneman
Sapiens	Yuval Noah

8 rows selected.

SQL> select l.genre as category from library l;

CATEGORY

fiction

fiction
fiction
fiction
fiction
science
non-fiction
non-fiction

8 rows selected.

Delete Command:

```
SQL> delete library where title = 'The Great Gatsby';
```

1 row deleted.

```
SQL> select * from library;
```

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

7 rows selected.

Arrange in Ascending order by Pages:

```
SQL> select title, pages, book_no from library order by pages;
```

TITLE	PAGES	BOOK_NO
War and Peace	142	103
WingsFire	210	106
The Guide	240	105
To Kill a Mockingbird	311	101
The Odyssey	420	104
ThinkFast	489	107
Sapiens	502	108

7 rows selected.

Arrange in Descending order by Pages for Fiction Books:

```
SQL> select title, pages, book_no, genre from library where genre = 'fiction' order by pages desc, book_no;
```

TITLE	PAGES	BOOK_NO	GENRE
The Odyssey	420	104	fiction
To Kill a Mockingbird	311	101	fiction
The Guide	240	105	fiction
War and Peace	142	103	fiction

Transaction Control Language(TCL):

Commit

SQL> select * from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

7 rows selected.

SQL> commit ;

Commit complete.

Save Point

Creating save point l2

SQL> select * from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

7 rows selected.

SQL> savepoint l2;

Savepoint created.

Creating savepoint s2

SQL> insert into library values(109, 'A Brief History of Time', 'Stephen Hawking', 'science', 256);

1 row created.

SQL> insert into library values(110, 'The Selfish Gene', 'Richard Dawkins', 'science', 360);

1 row created.


```
SQL> select * from library;
```

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502
109	A Brief History of Time	Stephen Hawking	science	256
110	The Selfish Gene	Richard Dawkins	science	360

9 rows selected.

```
SQL> savepoint e2;
```

Savepoint created.

Roll Back

Rollback to e2

```
SQL> insert into library values(111, 'Educated', 'Tara Westover', 'non-fiction', 334);
```

1 row created..

```
SQL> select * from library;
```

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502
109	A Brief History of Time	Stephen Hawking	science	256
110	The Selfish Gene	Richard Dawkins	science	360
111	Educated	Tara Westover	non-fiction	334

10 rows selected.

```
SQL> rollback to e2;
```

Rollback complete.

```
SQL> select * from library;
```

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502
109	A Brief History of Time	Stephen Hawking	science	256
110	The Selfish Gene	Richard Dawkins	science	360

9 rows selected.

Rollback to savepoint l2

SQL> rollback to l2;

Rollback complete.

SQL> select * from library;

BOOK_NO	TITLE	AUTHOR	GENRE	PAGES
101	To Kill a Mockingbird	-	fiction	311
103	War and Peace	Leo Tolstoy	fiction	142
104	The Odyssey	Homer	fiction	420
105	The Guide	R.K. Narayan	fiction	240
106	WingsFire	Kalam	science	210
107	ThinkFast	Kahneman	non-fiction	489
108	Sapiens	Yuval Noah	non-fiction	502

7 rows selected.

Commit

SQL> commit;

Commit complete.

CONTENTS	MARK ALLOTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
viva	10	
Total	60	

RESULT

Thus Data Definition Language commands and Integrity Constraints were executed